

Biostat Cplus 20 L - 3:1 vessel, 6-blade disc configuration (legacy)

sartorius-cplus-20

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	5.5 L	Source-supported
Maximum working volume	20 L	Source-supported
Nominal vessel / bag size	20 L	Source-supported
Total vessel volume loaded into the form	30 L	Source-supported
Initial working volume loaded into the form	5.5 L	Derived / estimated
Product-contact material	316L stainless steel / borosilicate glass / EPDM	Source-supported
Height : diameter	3	App assumption
Internal / chamber diameter	0.204 m	Derived / estimated
Liquid height at maximum working volume	0.612 m	Derived / estimated
Impeller / mixing type	Six-blade Rushton turbine	Source-supported
Impeller count	3	App assumption
Impeller diameter	0.0816 m	Derived / estimated
Power number	5	App assumption
Initial agitation	180 rpm	App assumption
Maximum agitation	1000 rpm	Source-supported
Initial aeration	0.5 vvm	App assumption
Maximum aeration	1.5 vvm	Source-supported
Baffled vessel	Yes	App assumption
Maximum pressure	1 bar(g)	App assumption
Default operating pressure	0.2 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 5.5 L

Source-supported | Sartorius Biostat Cplus brochure SBI1505-e - PDF page 13; Culture vessel, 20 L, 3:1 column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 5.5 L.

Limitations: Model and aspect-ratio column selected explicitly. Operating minimum differs from minimum fill for full sterilization (50% of maximum working volume).

Local original: [Sartorius Biostat Cplus brochure SBI1505-e - PDF page 13](#)

[Manufacturer publication](#)

Maximum working volume - 20 L

Source-supported | Sartorius Biostat Cplus brochure SBI1505-e - PDF page 13; Culture vessel, 20 L, 3:1 column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 20 L.

Limitations: Model and aspect-ratio column selected explicitly. Operating minimum differs from minimum fill for full sterilization (50% of maximum working volume).

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Nominal vessel / bag size - 20 L

Source-supported | Sartorius Biostat Cplus brochure SBI1505-e - PDF page 13; Culture vessel, 20 L, 3:1 column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 20 L.

Limitations: Model and aspect-ratio column selected explicitly. Operating minimum differs from minimum fill for full sterilization (50% of maximum working volume).

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Total vessel volume loaded into the form - 30 L

Source-supported | Sartorius Biostat Cplus brochure SBI1505-e - PDF page 13; Culture vessel, 20 L, 3:1 column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 30 L.

Limitations: Model and aspect-ratio column selected explicitly. Operating minimum differs from minimum fill for full sterilization (50% of maximum working volume).

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Initial working volume loaded into the form - 5.5 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: min(5.5 L, 20 L) = 5.5 L

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - 316L stainless steel / borosilicate glass / EPDM

Source-supported | Sartorius Biostat Cplus brochure SBI1505-e - PDF page 13; Culture vessel, 20 L, 3:1 column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 316L stainless steel / borosilicate glass / EPDM.

Limitations: Model and aspect-ratio column selected explicitly. Operating minimum differs from minimum fill for full sterilization (50% of maximum working volume).

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Manufacturer publication

Height : diameter - 3

App assumption | Sartorius Biostat Cplus brochure SBI1505-e - PDF page 13; Culture vessel H:D

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 3.

Limitations: Published ratio is vessel H/D, not liquid H/D. Using it for the liquid cylinder remains a model assumption.

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Manufacturer publication

Internal / chamber diameter - 0.204 m

Derived / estimated | Cylindrical geometry estimate

Rule: $D = \sqrt[3]{(4 \times V) / (\pi \times H:D)}$

Selection / calculation: $D = \sqrt[3]{(4 \times 0.02 \text{ m}^3) / (\pi \times 3)} = 0.2039888 \text{ m}$; the form displays 0.204 m.

Limitations: The estimate treats the liquid volume as a straight-sided cylinder and does not include heads, cones, internals or manufacturer-specific dimensions.

Liquid height at maximum working volume - 0.612 m

Derived / estimated | Cylindrical geometry estimate

Rule: liquid height = estimated internal diameter x H:D

Selection / calculation: $0.2039888 \text{ m} \times 3 = 0.6119663 \text{ m}$; the form displays 0.612 m.

Limitations: This is linked to the same ideal-cylinder approximation used for the diameter.

Impeller / mixing type - Six-blade Rushton turbine

Source-supported | Sartorius Biostat Cplus brochure SBI1505-e - PDF page 13; Culture vessel, 20 L, 3:1 column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Six-blade Rushton turbine.

Limitations: Model and aspect-ratio column selected explicitly. Operating minimum differs from minimum fill for full sterilization (50% of maximum working volume).

Local original: Sartorius Biostat Cplus brochure SBI1505-e - PDF page 13

Manufacturer publication

Impeller count - 3

App assumption | Generic reusable steel microbial profile

Rule: `PROFILE_DEFAULTS["steel-microbial"].impellerCount = 3`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. The count is used by the vessel diagram and represents a generic configuration unless source-supported.

Impeller diameter - 0.0816 m

Derived / estimated | Manufacturer diameter ratio applied to model geometry

Rule: $\text{impeller diameter} = \text{vessel diameter} \times 0.4$

Selection / calculation: $0.2039888 \times 0.4 = 0.08159551 \text{ m}$

Limitations: The ratio is published, but the vessel diameter is estimated. The resulting impeller diameter is therefore derived, not measured.

Local original: Sartorius Biostat Cplus brochure SBI1505-e - PDF page 13

Manufacturer publication

Power number - 5

App assumption | Generic reusable steel microbial profile

Rule: `PROFILE_DEFAULTS["steel-microbial"].powerNumber = 5`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. The power number is an empirical generic value used in $P = N_p N^3 D_i^5$. It is not recalculated from blade geometry.

Initial agitation - 180 rpm

App assumption | Generic reusable steel microbial profile

Rule: `PROFILE_DEFAULTS["steel-microbial"].baseRpm = 180`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This is the initial agitation setting loaded into the form.

Maximum agitation - 1000 rpm

Source-supported | Sartorius Biostat Cplus brochure SBI1505-e - PDF page 13; Culture vessel, 20 L, 3:1 column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1000 rpm.

Limitations: Model and aspect-ratio column selected explicitly. Operating minimum differs from minimum fill for full sterilization (50% of maximum working volume).

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Manufacturer publication

Initial aeration - 0.5 vvm

App assumption | Generic reusable steel microbial profile

Rule: `PROFILE_DEFAULTS["steel-microbial"].baseVvm = 0.5`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 1.5 vvm

Source-supported | Sartorius Biostat Cplus brochure SBI1505-e - PDF page 12; Gas supply, microbial application

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.5 vvm.

Limitations: Published maximum total gas rate for microbial application.

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Manufacturer publication

Baffled vessel - Yes

App assumption | Generic reusable steel microbial profile

Rule: PROFILE_DEFAULTS["steel-microbial"].baffled = true

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 1 bar(g)

App assumption | Generic reusable steel microbial profile

Rule: PROFILE_DEFAULTS["steel-microbial"].maxPressure = 1

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0.2 bar(g)

App assumption | Generic reusable steel microbial profile

Rule: PROFILE_DEFAULTS["steel-microbial"].operatingPressure = 0.2

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

Biostat Cplus is represented as a legacy/discontinued installed-base platform.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

Vessel design pressure: 3 bar(g) at 150 C; not substituted for the unverified process operating-pressure ceiling.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Sartorius Biostat Cplus brochure SBI1505-e

sartorius-biostat-cplus-brochure-sbi1505-e-bdf447cc4e44.pdf

SHA-256: bdf447cc4e4459d5702e9fb54eca9638d94665944c4d3f7154692b306157ecb1